

Problem set #8: Binomial Monte Carlo: Solutions

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Let the **volatility** of a stock be the standard deviation of its (continuously compounded) realized return on an annual basis. Then, we can define the up and down factors in the so-called *forward binomial tree* for a **non-dividend-paying** stock as

$$\begin{aligned}u &= e^{rh+\sigma\sqrt{h}} \\d &= e^{rh-\sigma\sqrt{h}}\end{aligned}\tag{1}$$

Let the continuously compounded, risk-free interest rate be 0.04.

```
r=0.04
```

Consider a non-dividend-paying stock whose current price is \$100 and whose volatility is 0.25. We will be pricing a one-year, at-the-money call option in a variety of ways here.

```
s0=100
sigma=0.25
T=1
K=s0
```

Problem #1: Analytic one period

Price the option above using a one period binomial tree.

Solution:

```
periods=1
h=T/periods
u=exp(r*h+sigma*sqrt(h))
d=exp(r*h-sigma*sqrt(h))
u
```

```
## [1] 1.336427
```

```
d
```

```
## [1] 0.8105842
```

```
#the payoff function
```

```
v.c<-function(x){
  max(x-K,0)
}
```

```
p.star=(exp(r*h)-d)/(u-d)
p.star
```

```
## [1] 0.4378235
```

```

#the possible stock prices
s.u=s0*u
s.d=s0*d

#the possible payoffs
v.u=v.c(s.u)
v.d=v.c(s.d)

v.0=exp(-r*T)*(p.star*v.u+(1-p.star)*v.d)
v.0

```

```
## [1] 14.15203
```

Problem #2: Monte Carlo one period

Price the option above using Monte Carlo a one period binomial tree. Use 10000 simulations.

Solution:

```

n.sims=10000
probs=c(p.star, 1-p.star)
factors=c(u,d)
final.prices=s0*sample(factors, size=n.sims, prob=probs, replace=TRUE)
#final.prices
payoffs<-pmax(final.prices-K,0)
#payoffs
v.bar=mean(payoffs)
#v.bar
v.0.mc<-exp(-r*T)*v.bar
v.0.mc

```

```
## [1] 14.1739
```

Problem #3: Analytic two periods

Price the above option using a two-period binomial tree.

Solution:

```

periods=2
h=T/periods
#h
u=exp(r*h+sigma*sqrt(h))
d=exp(r*h-sigma*sqrt(h))
#u
#d

p.star=(exp(r*h)-d)/(u-d)
#p.star

s.T=s0*c(u^2, u*d, d^2)
#s.T

v.T=pmax(s.T-K,0)
#v.T

probs=c(p.star^2, 2*p.star*(1-p.star), (1-p.star)^2)

```

```
v.c=exp(-r*T)*sum(v.T*probs)
v.c
```

```
## [1] 11.57622
```

Problem #4: Monte Carlo two periods

Price the option above using Monte Carlo using a two period binomial tree. Use 10000 simulations.

Solution:

```
#get the simulated values of the final stock price
#it's sufficient to simulate the number of upsteps
```

```
x=rbinom(n.sims, size=periods, prob=p.star)
```

```
x
```

```
## [1] 1 1 1 1 0 1 1 2 1 2 2 0 2 0 1 2 1 1 2 1 1 1 1 1 0 1 0 2 2 0 0 1 2 0 0 1
## [38] 1 1 0 2 1 2 0 2 1 1 1 2 0 1 2 1 1 0 1 1 1 1 1 1 1 0 1 1 0 1 2 2 1 1 1 2 0
## [75] 2 1 1 0 0 1 1 2 1 1 1 1 0 2 1 0 1 1 1 1 2 0 1 1 2 2
## [ reached 'max' / getOption("max.print") -- omitted 9900 entries ]
```

```
s.T=s0*u^x*d^(periods-x)
```

```
s.T
```

```
## [1] 104.08108 104.08108 104.08108 104.08108 73.08454 104.08108 104.08108
## [8] 148.22384 104.08108 148.22384 148.22384 73.08454 148.22384 73.08454
## [15] 104.08108 148.22384 104.08108 104.08108 148.22384 104.08108 104.08108
## [22] 104.08108 104.08108 104.08108 104.08108 73.08454 104.08108 73.08454
## [29] 148.22384 148.22384 73.08454 73.08454 104.08108 148.22384 73.08454
## [36] 73.08454 104.08108 104.08108 104.08108 73.08454 148.22384 104.08108
## [43] 148.22384 73.08454 148.22384 104.08108 104.08108 104.08108 148.22384
## [50] 73.08454 104.08108 148.22384 104.08108 104.08108 73.08454 104.08108
## [57] 104.08108 104.08108 104.08108 104.08108 104.08108 104.08108 73.08454
## [64] 104.08108 104.08108 73.08454 104.08108 148.22384 148.22384 104.08108
## [71] 104.08108 104.08108 148.22384 73.08454 148.22384 104.08108 104.08108
## [78] 73.08454 73.08454 104.08108 104.08108 148.22384 104.08108 104.08108
## [85] 104.08108 104.08108 73.08454 148.22384 104.08108 73.08454 104.08108
## [92] 104.08108 104.08108 104.08108 148.22384 73.08454 104.08108 104.08108
## [99] 148.22384 148.22384
## [ reached 'max' / getOption("max.print") -- omitted 9900 entries ]
```

```
#get the payoffs
```

```
v.T=pmax(s.T-K,0)
```

```
v.bar=mean(v.T)
```

```
v.c.mc=exp(-r*T)*v.bar
```

```
v.c.mc
```

```
## [1] 11.53741
```

Problem #5: Analytic one hundred periods

Price the above option using a 100-period binomial tree.

Solution:

```
periods=100
```

```
h=T/periods
```

```

#h
u=exp(r*h+sigma*sqrt(h))
d=exp(r*h-sigma*sqrt(h))
#u
#d

p.star=(exp(r*h)-d)/(u-d)
#p.star

k=periods:0
s.T=s0*u^k*d^(periods-k)
s.T

##      [1] 1267.967097 1206.127612 1147.304074 1091.349394 1038.123656 987.493768
##      [7] 939.333129 893.521311 849.943763 808.491516 769.060920 731.553376
##     [13] 695.875097 661.936868 629.653826 598.945247 569.734342 541.948071
##     [19] 515.516951 490.374893 466.459027 443.709552 422.069582 401.485005
##     [25] 381.904351 363.278656 345.561346 328.708121 312.676837 297.427407
##     [31] 282.921701 269.123447 255.998142 243.512965 231.636698 220.339643
##     [37] 209.593551 199.371553 189.648088 180.398842 171.600686 163.231622
##     [43] 155.270722 147.698079 140.494759 133.642749 127.124915 120.924960
##     [49] 115.027380 109.417428 104.081077 99.004983 94.176453 89.583414
##     [55] 85.214379 81.058425 77.105159 73.344696 69.767633 66.365025
##     [61] 63.128365 60.049558 57.120906 54.335087 51.685133 49.164420
##     [67] 46.766643 44.485807 42.316208 40.252422 38.289289 36.421898
##     [73] 34.645581 32.955896 31.348618 29.819728 28.365403 26.982006
##     [79] 25.666078 24.414328 23.223627 22.090998 21.013607 19.988761
##     [85] 19.013898 18.086579 17.204486 16.365414 15.567263 14.808039
##     [91] 14.085842 13.398867 12.745397 12.123797 11.532512 10.970065
##     [97] 10.435048 9.926125 9.442022 8.981529 8.543495

v.T=pmax(s.T-K, 0)

rn.probs=dbinom(k, size=periods, prob=p.star)

v.c=exp(-r*T)*sum(v.T*rn.probs)
v.c

## [1] 11.84362

```

Problem #6: Monte Carlo with one hundred periods

Price the option above using Monte Carlo with a hundred period binomial tree. Use 10000 simulations.

Solution:

```

#simulating the number of upsteps
x=rbinom(n.sims, size=periods, prob=p.star)
#x

#calculate the corresponding simulated stock prices
s.T=s0*u^x*d^(periods-x)
#hist(s.T, breaks=25)

#calculating the simulated payoffs
v.T=pmax(s.T-K,0)

```

```
#v.T

#average the simulated payoffs
v.bar=mean(v.T)
#v.bar

#discounting
v.O.mc=exp(-r*T)*v.bar
v.O.mc

## [1] 11.59166
```